

Drone Designer — Project Summary

What It Is

A **VB.NET WinForms application** (.NET Framework 4.7.2) that takes UAV mission requirements as input and recommends optimal drone hardware components (motors, batteries, ESCs, flight controllers, cameras, GPS, etc.).

File Tree

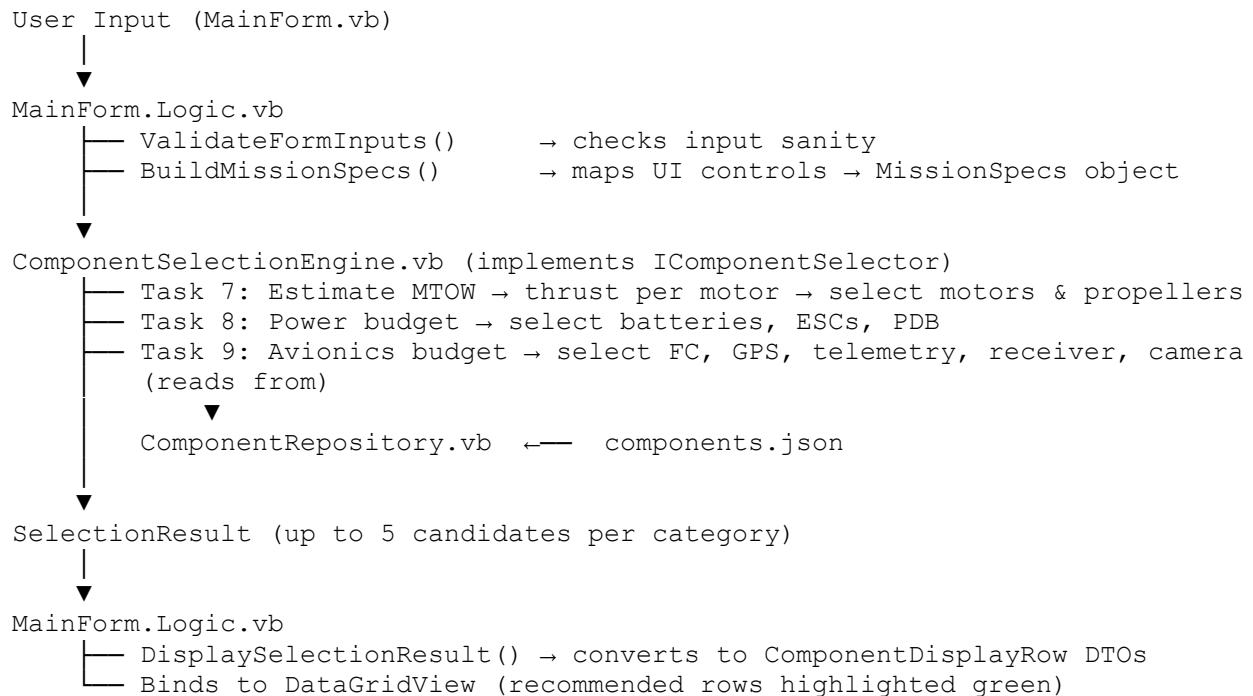
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Drone Designer/
├── Drone Designer.slnx                                # Solution file
├── Drone Designer.vbproj                             # MSBuild project (targets .NET 4.7.2,
WinExe)
├── App.config                                         # .NET startup config
├── Core/
│   ├── Data/
│   │   └── ComponentRepository.vb                    # Loads & indexes components.json;
typed queries
│   ├── Interfaces/
│   │   └── IComponentSelector.vb                    # Contract: SelectComponents(specs) →
result
│   ├── Models/
│   │   ├── ComponentSpecs.vb                        # 11 component subclasses + enums
│   │   │   (Motor, Battery, ESC...)
│   │   ├── MissionSpecs.vb                          # Mission input parameters (endurance,
range, payload...)
│   │   └── ComponentDisplayRow.vb                    # DTO mapping ComponentSpecs →
DataGridView columns
│   └── Services/
│       └── ComponentSelectionEngine.vb                # Core algorithm: Tasks 7→8→9
pipeline
├── UI/
│   ├── Forms/
│   │   ├── MainForm.vb                             # Root window layout (two tabs, all
controls in code)
│   │   └── MainForm.Logic.vb                         # Event handlers, validation, engine
wiring
│   └── MainForm.Designer.vb                          # Minimal auto-generated stub
Controls/                                             # (empty – reserved for future custom
controls)
├── Utilities/
│   └── ConfigManager.vb                             # Singleton config: paths, logging,
SolidWorks settings
├── Resources/
│   └── components.json                              # Component database (~1800 lines of
motors, ESCs, etc.)

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— Solidworks/	# (empty — reserved for Module 2 CAD
integration)	
— My Project/	# Auto-generated VB.NET project files
— bin/ & obj/	# Build output
— files.zip	# Archive

How the Files Connect (Data Flow)



Key File Roles

File	Role
ComponentSpecs.vb (1314 lines)	Data model backbone — 11 typed subclasses (MotorSpec, BatterySpec, etc.) with all hardware properties
MissionSpecs.vb (458 lines)	Input contract — flight performance, payload, environment, autonomy requirements
ComponentSelectionEngine.vb (1544 lines)	Brain — 3-stage pipeline with physics-based sizing (MTOW iteration, power budget, avionics budget)
ComponentRepository.vb (304 lines)	Data layer — deserializes JSON into typed objects, provides filtered queries
MainForm.vb (1023 lines)	UI shell — programmatically builds two-tab layout (input form + results grid)
MainForm.Logic.vb (492 lines)	Glue — validates input, builds MissionSpecs, calls engine, displays results

File	Role
ComponentDisplayRow.vb (236 lines)	View adapter — bridges model property names to DataGridView column names
ConfigManager.vb (313 lines)	Settings — paths, SolidWorks config, logging level
components.json (1829 lines)	Database — real drone hardware specs (T-Motor, Pixhawk, u-blox, etc.)